

Exam Date & Time: 26-Dec-2024 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech III Semester - Make up Examination - December - 2024

**Data Structures [ECM-2125]**

**Marks: 50**

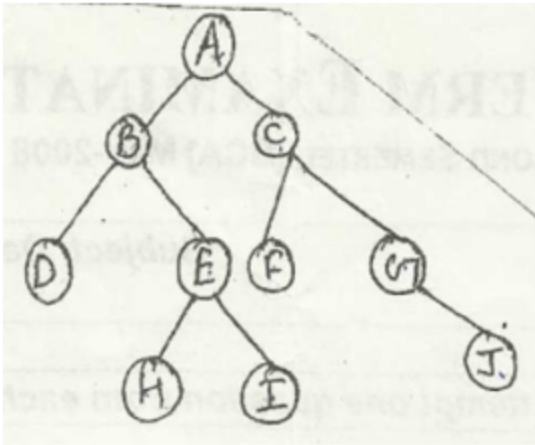
**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) What is data structure? What are various types of data structure? Explain (4)
  - A)
  - B) What is Call by Value and Call by Reference ? Write a c program and explain with example. (3)
  - C) Represent the given matrix in the form of array and linked list? What kind of matrix is it? Justify your answer. (3)
 

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 0 | 9 | 0 | 0 | 0 | 4 | 0 | 0 |
| 0 | 0 | 6 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 5 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 3 | 0 |
| 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 |
- 2) Write a program in 'C' to implement Stack using Linked-List. (4)
  - A)
  - B) Convert the infix expression into prefix expression.  
 $A B * C D + E / F / (G + H)$  (3)
  - C) Explain operations of a stack with an example (3)
- 3) List various operations of linked list and explain how to insert a node anywhere in the list (4)
  - A)
  - B) Write a complete C program consisting of a function sortedInsert(. . .) to insert an integer into a sorted doubly linked list, such that after insertion the list remains sorted. Also include function to display the contents of the list. (3)
  - C) What is a priority queue? Implement using a linked list. (3)

- 4) Write a function to check if a given binary tree is a binary search tree or not. (4)
- A)
- B) Construct a Binary Search Tree (BST) for in the order of the elements given as: 150, 130, 140, 180, 160, and 158,170. Explain the logic used to construct BST. Explain the steps to delete the following elements from the tree constructed. show the BST after each deletion of (4)
- 130
  - 180
  - 160
- C) What is a tree? With suitable example, define: (i) Binary Tree (ii) Level of Binary tree (iii) Complete Binary Tree (iv) Degree of the tree (2)
- 5) Find inorder, postorder and preorder traversal for the following tree and write recursive C program for preorder tree traversal.
- A)
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- (4)
- B) Write the algorithm for Depth First Traversal of graphs. (3)
- C) Construct a binary tree whose preorder and postorder traversals give the following sequence of vertices. (3)
- Preorder - ABCEIFJDGHKL
  - Postorder - IEJFCGKLHDBA

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